EMC First Midterm – Study Notes

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1.1 Chapter 1: Introduction to EMC

1.1.1 What is EMC?

Electromagnetic Compatibility (EMC) is the discipline of ensuring that electronic systems neither produce nor are affected by harmful electromagnetic interference (EMI). A system is EMC-compliant when it satisfies all three criteria simultaneously *(chap1.pdf, p.4)*:

1. It does **not cause** interference with other systems.
2. It is **not susceptible** to emissions from other systems.
3. It does **not cause** interference **with itself**.

Interference only occurs when unwanted coupling actually causes a receptor to **misbehave**. Mere coupling --- without degraded behaviour --- is not interference. *(chap1.pdf, p.5)*

1.1.2 Three-Stage EMC Coupling Model

Every EMC problem has three stages: **Source** → **Coupling Path** → **Receptor**. *(chap1.pdf, p.5)*

To fix an EMC problem, attack at one or more of these stages --- in order of cost-effectiveness *(chap1.pdf, p.6)*:

1. Suppress the **source** (cheapest, most effective).
2. Break or attenuate the **coupling path**.
3. Harden the **receptor**.

1.1.3 Four Basic EMC Subproblems *(chap1.pdf, p.7)*

#	Type	Path
1	Radiated emissions	Device → air → external
2	Radiated susceptibility	External → air → device
3	Conducted emissions	Device → power cord → system
4	Conducted susceptibility	System → power cord → device

The power cord carries not only 60 Hz but also high-frequency noise; these can radiate or conduct to other equipment. *(chap1.pdf, p.8)*

1.1.4 Electrical Dimensions and Waves

The key question is: how large is a circuit element or conductor relative to the wavelength of the highest-frequency signal?

$$k = \frac{L}{\lambda}$$

(chap1.pdf, p.21)

Condition	Appropriate model
$k \ll 0.1$ (i.e. $L < \lambda/10$)	Lumped-circuit theory (KVL, KCL)
$k \sim 1$ or larger	Maxwell's equations

Phase velocity in a medium with relative permittivity ε_r and permeability μ_r:

$$v = \frac{v_0}{\sqrt{\mu_r\,\varepsilon_r}}, \qquad v_0 = 3 \times 10^8 \text{ m/s}$$

(chap1.pdf, p.20)

Wavelength in the medium:

$$\lambda = \frac{v}{f} = \frac{v_0}{f\sqrt{\mu_r\,\varepsilon_r}}$$

(chap1.pdf, p.21)

A denser medium (larger ε_r) $\rightarrow$ smaller λ $\rightarrow$ larger electrical dimension $\rightarrow$ the circuit is **more like a transmission line** and less like a lumped element.

Phase difference over a length L:

$$\Delta\phi = \beta L = \frac{2\pi L}{\lambda}$$

(chap1.pdf, p.17)

This phase shift is equivalent to a **time delay** $T_D = L/v$. *(chap1.pdf, p.16--19)*

1.1.5 Decibels and EMC Units

dB Definitions

$$G_{p,\mathrm{dB}} = 10\log_{10}\left(\frac{P_{out}}{P_{in}}\right)$$

(chap1.pdf, p.25)

$$G_{V,\mathrm{dB}} = 20\log_{10}\left(\frac{V_{out}}{V_{in}}\right)$$

(chap1.pdf, p.26)

When input and load resistances are equal ($R_{in} = R_L$), the power gain equals the voltage gain in dB. *(chap1.pdf, p.26)*

dB Arithmetic Gains add, losses subtract in dB:

$$P_{out,\mathrm{dBm}} = G_{\mathrm{dB}} + P_{in,\mathrm{dBm}}$$

(chap1.pdf, p.28)

Cable Attenuation For a matched, lossy cable (attenuation constant α Np/m), power falls off exponentially. The attenuation in dB per unit length is:

$$\alpha\,[\mathrm{Np/m}] = \frac{P_L\,[\mathrm{dB/m}]}{8.686}$$

(chap1.pdf, p.30)

Voltage--Power relationship (50 Ω system):

$$\mathrm{dB}\mu\mathrm{V}_{\mathrm{RMS}} = 107 + \mathrm{dBm}$$

(chap1.pdf, p.30)

50 Ω Standard Impedance *(chap1.pdf, p.33)* All modern EMC test equipment (spectrum analysers, signal sources, cables) uses 50 Ω. When the load equals the cable characteristic impedance, the input impedance is constant at all frequencies and cable lengths --- essential for swept-frequency measurements.

1.2 Chapter 2: EMC Regulatory Requirements

1.2.1 Overview

Two categories of requirements *(chap2.pdf, p.3)*: - **Governmental** (FCC, CISPR, MIL-STD): legally mandatory. - **Manufacturer-imposed**: for product quality/reliability.

1.2.2 FCC (USA)

Any digital device using a clock > 9 kHz must comply with FCC Part 15. *(chap2.pdf, p.5)*

Class	Environment	Limits
A	Commercial/industrial	Less stringent
B	Residential	More stringent

Conducted emissions: 150 kHz -- 30 MHz, measured with a LISN. *(chap2.pdf, p.7)*
Radiated emissions: 30 MHz -- 40 GHz, measured at OATS or SAC. *(chap2.pdf, p.7)*
Class A measured at 10 m; Class B at 3 m. After normalising Class A to 3 m:

$$\Delta = 20\log_{10}\left(\frac{10}{3}\right) \approx 10.46 \text{ dB}$$

(chap2.pdf, p.13)

Class A limits are ˜10 dB **less stringent** than Class B. *(chap2.pdf, p.13)*

1.2.3 CISPR 22 (International)

- Conducted limits: same as FCC. *(chap2.pdf, p.15)*
- Radiated limits: up to 1 GHz only; measured at 10 m. *(chap2.pdf, p.16)*
- First standard to mandate **susceptibility (immunity)** testing for commercial products. *(chap2.pdf, p.16)*

1.2.4 MIL-STD-461E

Much wider frequency range; requirements can be waived or tailored. Key tests: RS103 (radiated susceptibility), CE102 (conducted emissions), RE102 (radiated emissions). *(chap2.pdf, pp.17--23)*

1.2.5 Measurement Infrastructure

Semianechoic Chamber (SAC) ***(chap2.pdf, pp.25--27)***

- Top + sides: RF absorbers; Floor: ground plane.
- Measurement antenna scanned from 1--4 m height; maximum emission recorded.
- Detector: quasi-peak (FCC/CISPR), peak (MIL).
- Antennas: biconical (30--200 MHz), log-periodic (200 MHz -- 1 GHz), rod dipole/horn (MIL).

LISN ***(chap2.pdf, pp.28--32)*** The Line Impedance Stabilisation Network (LISN): - Presents a **constant 50** Ω to the EUT at RF - Rejects 60 Hz power, passes 150 kHz -- 30 MHz noise to the receiver

A PCB that complied with the DIP-oscillator design still exceeded Class B by 30 dB (horizontal) and 15 dB (vertical) because the board itself --- $\lambda/2$ at 1 GHz --- acted as an antenna. *(chap2.pdf, p.38)*

1.2.6 EMC Design Principles *(chap2.pdf, p.41)*

"Paper work before layout" --- analyse EMC before committing to PCB layout.

"Prepare holes for capacitors and resistors" --- reserve component footprints for EMC fixes.

1.3 Chapter 3: Signal Spectra

1.3.1 Why Spectra Matter

EMC compliance depends on the **frequency content** of signals, not just their amplitude or fundamental frequency. The **rise/fall time** dominates high-frequency spectral content. *(chap3.pdf, p.3)*

1.3.2 Fourier Series of a Periodic Signal

$$x(t) = \sum_{n=-\infty}^{\infty} c_n \, e^{jn\omega_0 t}, \qquad \omega_0 = 2\pi/T$$

(chap3.pdf, p.7)

Expansion coefficients:

$$c_n = \frac{1}{T}\int_0^T x(t)\, e^{-jn\omega_0 t}\, dt, \qquad c_0 = \frac{1}{T}\int_0^T x(t)\, dt$$

(chap3.pdf, p.8)

One-sided spectrum: $c_n^+ = 2|c_n|$ for $n > 0$; $c_0^+ = c_0$. *(chap3.pdf, p.9)*

1.3.3 Square-Wave Pulse Train *(chap3.pdf, p.11)*

Amplitude A, pulse width τ, period T, duty cycle $D = \tau/T$:

$$c_n = AD\,\frac{\sin(n\pi D)}{n\pi D}$$

(chap3.pdf, p.11)

This is a **sinc function** in n. At $D = 1/2$, even harmonics vanish. *(chap3.pdf, p.14)*

1.3.4 Computational Shortcuts *(chap3.pdf, pp.19--24)*

Property	Effect on c_n
Linearity	c_n scales and adds
Time-shift by α	Multiply by $e^{-jn\omega_0\alpha}$
Derivative	Multiply by $(jn\omega_0)^k$
Impulse train $\sum \delta(t - kT)$	All $c_n = 1/T$

Practical method: differentiate until impulses appear, find their c_n, then divide by $(jn\omega_0)^k$.

1.3.5 Trapezoidal Clock Waveform *(chap3.pdf, pp.27--30)*

With rise time $\tau_r = \tau_f$:

$$|c_n| = AD\left|\frac{\sin(n\pi D)}{n\pi D}\right| \cdot \left|\frac{\sin(n\pi f_0\tau_r)}{n\pi f_0\tau_r}\right|$$

(chap3.pdf, p.29)

Product of two sinc functions --- one for duty cycle, one for rise time.

1.3.6 Spectral Bounds (Bode Envelope) *(chap3.pdf, pp.33--34)*

The envelope of $2|c_n|$ vs. frequency:

Region	Behavior
$f < 1/(\pi\tau)$	Flat at $2AD$
$1/(\pi\tau) < f < 1/(\pi\tau_r)$	-20 dB/decade
$f > 1/(\pi\tau_r)$	-40 dB/decade

Bandwidth:

$$BW \approx \frac{1}{\tau_r}$$

(chap3.pdf, p.37)

Key takeaway: To reduce high-frequency emissions, **slow down the edges** (increase τ_r). Reducing duty cycle lowers low-frequency content but does not affect high-frequency content. *(chap3.pdf, p.35, p.39)*

1.3.7 Ringing *(chap3.pdf, pp.40--42)*

Undershoot/overshoot (from PCB trace L and C) increases emissions at the **ringing frequency**. Remedies: series damping resistor, ferrite bead, or matched transmission line.

1.3.8 Spectrum Analyser *(chap3.pdf, pp.44--46)*

- **Resolution bandwidth (RBW):** 6 dB bandwidth. Use smallest RBW for lowest noise floor.
- Regulatory measurements use the **quasi-peak (QP)** detector.
- Harmonic spacing must exceed RBW to avoid blending.

1.3.9 Fourier Transform *(chap3.pdf, p.47)*

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt, \qquad x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t}\, d\omega$$

(chap3.pdf, p.47)

Linear system output: $Y(j\omega) = H(j\omega)\, X(j\omega)$. *(chap3.pdf, p.49)*

1.3.10 Random (PCM-NRZ) Signals *(chap3.pdf, pp.50--53)*

Power spectral density via Wiener-Kinchine theorem:

$$S_{xx}(f) = \mathcal{F}\{R_{xx}(\tau)\}$$

(chap3.pdf, p.52)

For PCM-NRZ (bit period T, levels 0 and X_0):

$$S_{xx}(f) = \frac{X_0^2\, T}{4}\, \mathrm{sinc}^2(fT)$$

(chap3.pdf, p.52)

This is equivalent to squaring the spectrum of a 50% duty-cycle square pulse train.

1.4 Chapter 4: Transmission Lines and Signal Integrity

1.4.1 Why Transmission Lines?

When a conductor's physical length approaches $\lambda/10$ at the highest signal frequency, lumped-circuit theory fails and the conductor must be treated as a **transmission line**. The line imposes a time delay and can cause reflections. *(chap4.pdf, p.3)*

1.4.2 Lossless Line Equations *(chap4.pdf, pp.8--10)*

KVL and KCL on a Δz section give:

$$-\frac{\partial V}{\partial z} = l\,\frac{\partial I}{\partial t}, \qquad -\frac{\partial I}{\partial z} = c\,\frac{\partial V}{\partial t}$$

(chap4.pdf, pp.8--9)

Decoupled wave equations:

$$\frac{\partial^2 V}{\partial z^2} = lc\,\frac{\partial^2 V}{\partial t^2}, \qquad \frac{\partial^2 I}{\partial z^2} = lc\,\frac{\partial^2 I}{\partial t^2}$$

(chap4.pdf, p.10)

1.4.3 TEM Mode Relationship *(chap4.pdf, p.12)*

For homogeneous surrounding medium:

$$lc = \mu\varepsilon, \qquad v = \frac{1}{\sqrt{lc}} = \frac{v_0}{\sqrt{\mu_r\, \varepsilon_r}}$$

(chap4.pdf, p.12)

This allows l and c to be computed using static (dc) field solutions. *(chap4.pdf, p.12)*

1.4.4 Per-Unit-Length Parameters (Key Results)

Structure	l [H/m]	c [F/m]
Two-wire (exact, radii a, sep. d)	$\frac{\mu_0}{\pi}\cosh^{-1}\frac{d}{2a}$	$\frac{\pi\varepsilon_0}{\cosh^{-1}(d/2a)}$
Wire above ground (height h, radius a)	$\frac{\mu_0}{2\pi}\cosh^{-1}\frac{h}{a}$	$\frac{2\pi\varepsilon_0}{\cosh^{-1}(h/a)}$
Coaxial (inner a, outer b)	$\frac{\mu_0}{2\pi}\ln\frac{b}{a}$	$\frac{2\pi\varepsilon_0\varepsilon_r}{\ln(b/a)}$

(chap4.pdf, pp.19--26)

Note: $lc = \mu_0\varepsilon_0$ for all lossless homogeneous-medium lines. *(chap4.pdf, p.12)*

1.4.5 Time-Domain Solution *(chap4.pdf, pp.34--37)*

General solution:

$$V(z,t) = V^+(t - z/v) + V^-(t + z/v)$$

(chap4.pdf, p.34)

$$I(z,t) = \frac{V^+(t - z/v) - V^-(t + z/v)}{Z_C}, \qquad Z_C = \sqrt{l/c}$$

(chap4.pdf, p.34)

Load reflection coefficient:

$$\Gamma_L = \frac{Z_L - Z_C}{Z_L + Z_C}$$

(chap4.pdf, p.35)

Source reflection coefficient (identical form):

$$\Gamma_S = \frac{Z_S - Z_C}{Z_S + Z_C}$$

(chap4.pdf, p.36)

Initial forward voltage (launched at $t = 0$):

$$V^+_{\text{init}} = V_S\,\frac{Z_C}{Z_S + Z_C}$$

(chap4.pdf, p.37)

The current reflection coefficient is **negative** of the voltage reflection coefficient. *(chap4.pdf, p.35)*

1.4.6 Bounce Diagram *(chap4.pdf, pp.42--43)*

Plot of position z vs. time t; each line segment is a traveling wave. Used to determine $V(z_0, t)$ or $V(z, t_0)$ graphically by summing all waves that have reached the point.

When both ends are matched ($\Gamma_S = \Gamma_L = 0$): only a time delay $T_D = L/v$ exists.

1.4.7 Signal Integrity Criteria *(chap4.pdf, p.59)*

The line can be treated as a lumped element (negligible distortion) when:

$$\tau_r > 10\,T_D \quad \text{or equivalently} \quad L < \frac{\lambda}{10}$$

(chap4.pdf, p.59)

Simulation results: at $\tau_r = T_D$, overshoot is 7 V for a 5 V supply; at $\tau_r = 10\,T_D$, it drops to 5.3 V. *(chap4.pdf, pp.60--63)*

1.4.8 Termination Loads *(chap4.pdf, pp.50--53)*

Capacitive load (source-matched, $Z_S = Z_C$):

$$V_L(t) = V_0\Big(1 - e^{-t/\tau_C}\Big)\, u(t - T_D), \qquad \tau_C = Z_C\, C_L$$

(chap4.pdf, pp.50--52)

Initial response: capacitor looks like a short circuit; final: open circuit. Added 50% delay: $0.693\, Z_C C_L$.

Inductive load: opposite behaviour --- initially open, finally short:

$$\tau_L = L_L / Z_C$$

(chap4.pdf, p.53)

1.4.9 Matching Schemes *(chap4.pdf, pp.54--55)*

Series match (add R at source so $Z_S + R = Z_C$): - Load voltage = full source voltage. *(chap4.pdf, p.54)* - No DC power loss. *(chap4.pdf, p.54)*

Parallel match (add R across load so $R = Z_C$ for high-Z load): - Load voltage is less than source voltage. *(chap4.pdf, p.55)* - Consumes power; fails at high frequencies with capacitive loads. *(chap4.pdf, pp.55, 58)*

For lines with discontinuities: must use **series match at source AND parallel match at load** to eliminate all reflections. *(chap4.pdf, p.67)*

1.4.10 Discontinuity Coefficients *(chap4.pdf, pp.64--65)*

At a junction between Z_{C1} and Z_{C2}:

$$\Gamma_{12} = \frac{Z_{C2} - Z_{C1}}{Z_{C2} + Z_{C1}}, \qquad T_{12} = 1 + \Gamma_{12}$$

(chap4.pdf, pp.64--65)

1.4.11 Phasor (Sinusoidal) Solution *(chap4.pdf, pp.73--78)*

$$\hat{V}(z) = V^+\, e^{-j\beta z} + V^-\, e^{+j\beta z}, \qquad \beta = \omega\sqrt{lc}$$

(chap4.pdf, p.73)

Input impedance (distance d from load):

$$Z_{in}(-d) = Z_C\, \frac{Z_L + jZ_C \tan(\beta d)}{Z_C + jZ_L \tan(\beta d)}$$

(chap4.pdf, p.75)

VSWR:

$$S = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|}$$

(chap4.pdf, p.77)

Average power (lossless line):

$$P_{av} = \frac{|V^+|^2}{2Z_C}\big(1 - |\Gamma_L|^2\big)$$

(chap4.pdf, p.78)

1.4.12 Lossy Line *(chap4.pdf, pp.79--81)*

Propagation constant:

$$\gamma = \sqrt{(r + j\omega l)(g + j\omega c)} = \alpha + j\beta$$

(chap4.pdf, p.80)

Both α (attenuation) and β (phase) are frequency-dependent $\rightarrow$ dispersion.

1.4.13 Conductor and Dielectric Losses *(chap4.pdf, pp.82--87)*

Loss mechanism	Resistance / conductance	Frequency dependence
Skin effect (circular wire)	$r_{hf} = \frac{1}{2\pi r_w \sigma \delta}$	$\propto \sqrt{f}$ (10 dB/dec)
Skin effect (PCB land $w \times t$)	$r_{hf} = \frac{1}{2\sigma w \delta}$	$\propto \sqrt{f}$
Dielectric loss	$g = \omega c \tan\theta$	$\propto f$ (20 dB/dec)

Skin depth: $\delta = 1/\sqrt{\pi f \mu \sigma}$ --- decreases with $\sqrt{f}$. *(chap4.pdf, p.82)*
For FR-4: $\tan\theta \approx 0.02$. *(chap4.pdf, p.94)*

1.4.14 Low-Loss Approximation *(chap4.pdf, pp.90--91)*

Valid when $r \ll \omega l$ and $g \ll \omega c$ (above ˜5 MHz for typical PCB):

$$\alpha \approx \frac{r}{2Z_C} + \frac{gZ_C}{2}, \qquad \beta \approx \omega\sqrt{lc}, \qquad Z_C \approx \sqrt{l/c}$$

(chap4.pdf, p.91)
Above 20 MHz: no β dispersion; α dispersion from skin effect remains.

1.4.15 Lumped-Circuit Model Validity *(chap4.pdf, pp.95--97)*

Use lumped models only when $L < \lambda/10$.

Model	Valid frequency range
Lumped-π, 1 section	Up to 10 MHz (example: 7-cm line on FR-4)
Lumped-π, 2 sections	Up to 20 MHz

Each additional section extends the range by approximately the same factor.